

7. Region Splitting

Pixel set: [40, 45, 100, 105]

(a) Interpretation of the segmentation problem

The pixel set contains two visibly different intensity clusters (around 40–45 and around 100–105). Treating this set as one homogeneous region would be inaccurate, since the internal variation is too large — the region needs to be split into smaller, more uniform sub-regions.

(b) Justification for the splitting method

The split-and-merge technique recursively divides a region into quadrants/subsets whenever the intensity variation (e.g. max – min, or variance) within it exceeds a chosen limit, continuing until each resulting region is sufficiently homogeneous.

(c) Apply threshold = 80

Rule: pixels < 80 are grouped into Region 1; pixels ≥ 80 are grouped into Region 2.

(d) Computed regions

Region	Pixel values	Mean
Region 1	40, 45	42.5
Region 2	100, 105	102.5

(e) Interpretation of separation

Splitting at threshold 80 cleanly separates the pixel set into two homogeneous regions with a large gap between their means (42.5 vs. 102.5), confirming they represent two genuinely distinct intensity clusters — for example, two different objects or an object and background.